

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

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ABSTRACT

Representation alignment (REPA) accelerates generative training by distilling the patch-level representations of a strong, pretrained vision encoder into the intermediate features of a diffusion transformer. Despite its practical success, a foundational question has remained unresolved: which property of the target representation actually governs generation quality—its *global* semantic content (for instance, as summarized by ImageNet-1K accuracy) or the spatial structure of its patch tokens (the pairwise cosine similarity among those tokens)? The prevailing assumption holds that stronger global semantic performance yields a better alignment target. Through a large-scale empirical analysis spanning 27 vision encoders and three model scales, we show that this assumption is at best weak and often misleading: across encoders, it is spatial structure rather than global performance that predicts a target representation’s generative utility, correlating far more strongly with generation quality (Pearson $|r| > 0.85$) than ImageNet-1K accuracy does ($|r| = 0.26$). Building directly on this finding, we introduce two minimal modifications that specifically amplify the transfer of *spatial* information: we replace REPA’s standard three-layer MLP projection with a single convolutional layer, and we apply a lightweight spatial normalization to the target patch features. Implemented in fewer than four lines of code, the resulting method—termed iREPA—consistently accelerates the convergence of REPA, and the improvement is systematic across a diverse range of vision encoders, model sizes, alignment depths, and training recipes (REPA, REPA-E, MeanFlow, and JiT), persisting under classifier-free guidance and at longer training budgets. Together, these results identify spatial structure as the property that most reliably accounts for the effectiveness of representation alignment in the diffusion-transformer settings we study, and motivate a principled reconsideration of how alignment targets are selected, measured, and leveraged to train generative models more efficiently.

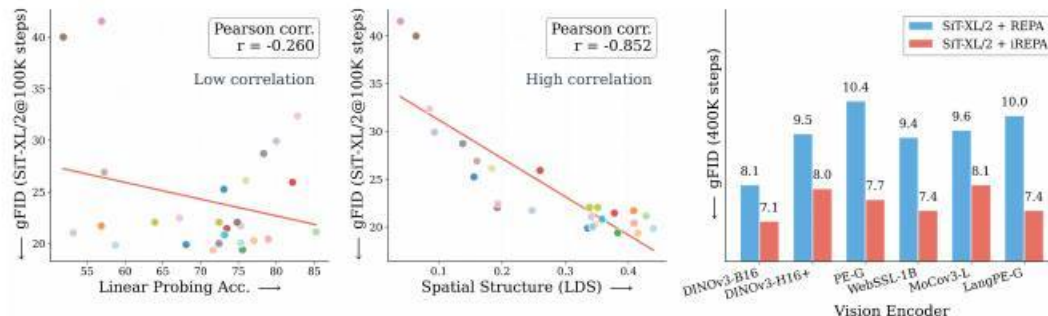


Figure 1: **What matters for representation alignment?** *Left:* Correlation analysis across 27 diverse vision encoders. Contrary to the prevailing wisdom, it is spatial structure—not global performance (measured by linear probing accuracy)—that best predicts a target representation’s generation performance. *Right:* Acting on this finding, we introduce two minimal modifications that accentuate the transfer of spatial features from the target representation to the diffusion model; the resulting method (iREPA) consistently accelerates the convergence of REPA across the diverse settings we evaluate.



Figure 2: **Motivating examples — spatial structure matters.** Comparing metrics across encoders reveals an inverse relationship between ImageNet accuracy and generation quality under REPA. *Left:* despite its much higher validation accuracy (82.8% vs. 53.1%), PE-G generates *worse* than SpatialPE-B (Bolya et al., 2025) (FID 32.3 vs. 21.0). *Right:* likewise, WebSSL-1B (Fan et al., 2025) attains far higher global performance (76.0% vs. 53.1%) yet worse generation. **Spatial Self-Similarity:** spatial structure, by contrast, tracks generation quality in both comparisons and resolves both reversals (see §3 for the spatial structure metric). All results reported at 100K using SiT-XL/2 and REPA.

1 INTRODUCTION

Representation alignment has emerged as a powerful and widely adopted technique for accelerating the training of diffusion transformers (Ma et al., 2024; Peebles & Xie, 2023). By regularizing the internal representations of a diffusion model toward those of a pretrained self-supervised visual encoder, recent methods deliver substantial gains in both convergence speed and final generation quality (Yu et al., 2024; Leng et al., 2025a). Yet these empirical successes have outpaced our understanding of *why* they occur: the property of the target representation that drives the acceleration has remained largely uncharacterized. This gap raises a foundational question. Is the benefit governed principally by richer global semantic information—the quantity conventionally summarized through linear probing accuracy—or does it instead arise from the spatial structure of the target representation, that is, the organization of relationships among its patch-token representations? Because the answer dictates how practitioners should select an alignment target, resolving it is of immediate practical consequence; across the encoders and diffusion-transformer settings we study, our evidence points clearly to the latter.

Understanding which property drives alignment is central to advancing generative training, because it directly determines how practitioners select an alignment target and how they diagnose whether alignment is succeeding. The prevailing view ties an encoder’s alignment quality to its ImageNet-1K validation accuracy, treated as a proxy for global semantic understanding (Oquab et al., 2024; Chen et al., 2021): target representations with stronger ImageNet performance are expected to yield better generation (Yu et al., 2024). Under this account, increases in the linear probing accuracy of diffusion features are routinely cited as evidence that representation alignment is working, which further entrenches global semantic information as the presumed primary driver of improvement. The following passage from REPA (Yu et al., 2024) crystallizes this widely held understanding:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

In this paper, we subject this conventional wisdom to a systematic empirical test, contrasting the two candidate explanations—global semantic information versus spatial structure—as predictors of what drives representation alignment. Our analysis spans a broad and representative range of 27 vision encoders, including recent large-scale vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perceptual encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024). It yields three findings that, taken together, reframe how representation alignment for diffusion models should be understood.

Higher validation accuracy does not imply a better representation for generation. Contrary to the prevailing assumption, vision encoders with stronger global semantic performance—as measured by ImageNet-1K linear probing accuracy—frequently underperform as alignment targets. Consider PE-Spatial-B, a spatially tuned model derived from PE-Core-G (Bolya et al., 2025). Despite a

substantially lower patch-token validation accuracy¹ than PE-Core-G (53.1% vs. 82.8%), PE-Spatial-B nonetheless produces better generation under REPA (Figure 2). WebSSL-1B (Fan et al., 2025) displays the same inversion: markedly higher global performance (76.0% vs. 53.1%), yet weaker generation. As we establish in Section §2, such inversions are not isolated anecdotes but recur systematically across target representations, pointing to a regularity in how representation alignment benefits diffusion training that the accuracy-centric view fails to capture.

Spatial structure, not global information, is the reliable predictor of generation performance.

To render this claim quantitative, we introduce a family of straightforward, fast-to-compute metrics that quantify the spatial self-similarity structure (Shechtman & Irani, 2007) among patch tokens (§3), and we conduct a large-scale correlation analysis relating REPA generation FID to both linear probing accuracy and spatial structure across 27 vision encoders and three model sizes (SiT-B, SiT-L, SiT-XL). The result is consistent and pronounced: every spatial structure metric correlates strongly with generation FID (Pearson $|r| > 0.852$), far surpassing the weak predictive power of ImageNet-1K validation accuracy ($|r| = 0.26$). Controlled interventions corroborate this pattern, showing that representations carrying very little global information already deliver substantial REPA gains. SAM2, for instance, yields better generation under REPA than encoders with roughly 60% higher ImageNet accuracy (Fig. 3); and even classical, purely spatial descriptors such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005)—though not the strongest targets—achieve meaningful gains as alignment targets (§3).

Accentuating spatial features improves convergence. Having identified spatial structure as the property that best predicts alignment quality, we translate this principle into two minimal modifications (fewer than four lines of code) that specifically amplify the transfer of *spatial* information from the target representation to the diffusion model. (1) A spatial normalization layer that sharpens the spatial contrast of the target representation. (2) A convolutional projection layer that replaces the standard MLP: we observe that the MLP projection—used to map diffusion features onto the target representation dimensionality—attenuates spatial contrast (Figure 6a), whereas a single convolution layer preserves it. The resulting method, termed iREPA, consistently accelerates the convergence of REPA across encoders, model sizes, and training recipes, including REPA-E (Leng et al., 2025a) and MeanFlow (Geng et al., 2025) with REPA, at negligible parameter and compute overhead.

Our main contributions are as follows:

- **An empirical account of what governs representation alignment.** Through a large-scale analysis over 27 vision encoders and three model scales, we establish that the spatial structure of a target representation—rather than its global semantic content—is the property that most reliably predicts alignment quality, revising the prevailing ImageNet-accuracy-centric view.
- **A predictive Spatial Structure Metric (SSM).** We introduce a straightforward, fast-to-compute metric that quantifies the spatial self-similarity of a target representation and, on the encoders we study, correlates far more strongly with downstream FID than linear probing accuracy—offering a concrete, training-free criterion for comparing candidate alignment targets.
- **iREPA, a principled and broadly applicable improvement to REPA.** Guided by this account, we propose two minimal modifications that amplify spatial-information transfer from the target representation to the diffusion features. The resulting method delivers consistent convergence-speed gains over REPA across variations in target representation, model architecture, alignment depth, and training recipe (REPA, REPA-E, MeanFlow with REPA, and JiT with REPA), at negligible added cost.

2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

We begin with a series of motivating observations that jointly make a single point: higher global performance (ImageNet-1K accuracy) in a target representation does not, on the encoders we examine, translate into better generation with REPA. Each of these observations resists explanation under the conventional, accuracy-centric view; in §3 we show that measuring the spatial structure of the target representation accounts for all of them within one coherent picture.

¹Following REPA (Yu et al., 2024), we measure the global semantic performance of an external representation via linear probing accuracy on patch tokens, since only patch tokens are used for representation alignment.

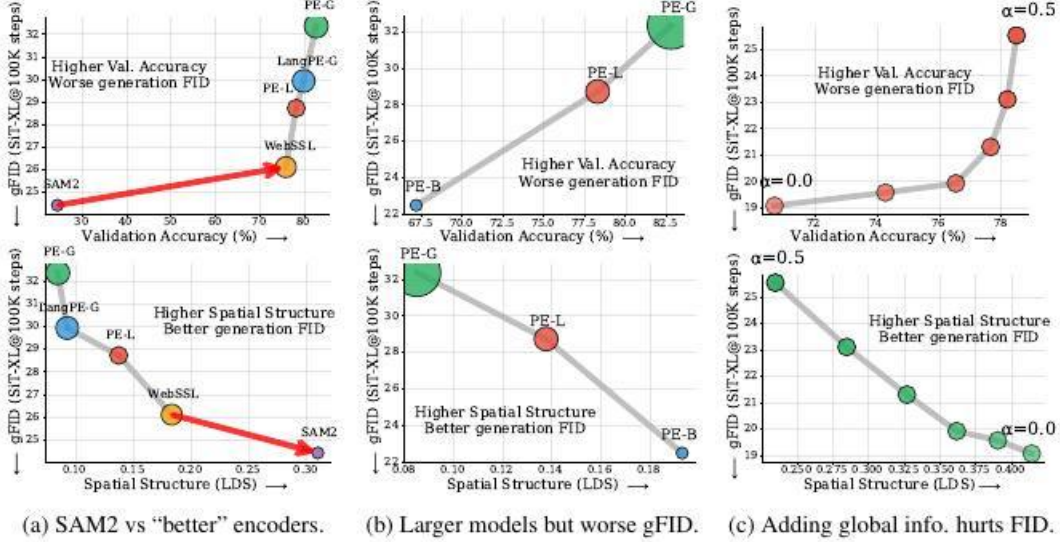


Figure 3: **Higher global information does not imply better REPA performance.** **Top row:** three trends in which global performance fails to track generation FID under REPA. (a) SAM2-S, with a validation accuracy of only 24.7%, yields better generation under REPA than models with $\sim 60\%$ higher validation accuracy. (b) Larger encoders within the same family can attain higher validation accuracy yet worse generation. (c) Adding global information to patch tokens via the CLS token improves global performance but degrades generation. **Bottom row:** across all three cases, spatial structure—rather than global performance—is the reliable indicator of generation quality. See §3 for the large-scale analysis across the different spatial structure metrics. All results are reported w/o classifier free guidance, SiT-XL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

Recent vision encoders. We examine a representative set of recent vision encoders spanning distinct training paradigms—Perceptual Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025). Consider PE-Spatial-B (80M), a compact spatially tuned model derived from PE-Core-G (1.88B) (Bolya et al., 2025). As Figure 2 shows, PE-Core-G attains far higher ImageNet-1K accuracy (82.8% vs. 53.1%) yet performs distinctly worse as a REPA target representation (FID 32.3 vs. 21.0). WebSSL-1B (1.2B) reproduces the same reversal: substantially higher ImageNet-1K accuracy (76.0% vs. 53.1%), but worse generation when used as the target representation for REPA (FID 26.1 vs. 21.0). In both comparisons, the encoder with the weaker global representation is the stronger alignment target—the opposite of what the accuracy-centric view predicts.

SAM2 outperforms vision encoders with much higher ImageNet-1K accuracy. To probe how little global information is in fact required, we analyze the SAM2-S vision encoder (46M) (Ravi et al., 2024), a compact model that carries minimal global information, with a validation accuracy of only 24.1% (Fig. 3a). Despite this, SAM2-S is a stronger REPA target than encoders with vastly higher ImageNet-1K accuracy, achieving a better FID than, *e.g.*, PE-Core-G (82.8%). Generative utility therefore persists even when global semantic content is nearly absent, indicating that such content is not a prerequisite for an effective alignment target.

Larger models within the same encoder family can yield similar or worse generation. Scaling up within an encoder family is widely presumed to produce better representations, as reflected in rising ImageNet-1K accuracy. For representation alignment, however, larger variants often deliver comparable (DINOv2) or even *degraded* (PE, C-RADIO) generation performance (Fig. 3b). Yu et al. (2024) report a related observation for DINOv2, attributing it to the conjecture that “all DINOv2 models being distilled from the DINOv2-g model and thus sharing similar representations.” In §3 we offer a more complete and unifying account grounded in spatial structure, one that subsumes this special case.

Adding more global information can hurt generation. To test the assumption directly, we perform a controlled intervention that injects global semantics—carried by the CLS token—into the local patch tokens, using DINOv2 as the encoder. The mixing operation is $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, where $\mathbf{p}_i$ denotes the i^{th} patch token, $\mathbf{c}$ the CLS token, and $\alpha \in [0, 0.5]$ controls the mixing strength. As reported in Figure 3c, increasing α from 0 to 0.5 raises linear probing accuracy monotonically from

70.7% to 78.5%, yet generation quality moves in the opposite direction, with FID deteriorating from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$. Deliberately adding global information thus *degrades* generation even as it improves the standard proxy for representation quality—a direct, controlled counterexample to the accuracy-centric view.

Taken together, these four observations converge on a single conclusion: within the settings we study, the global performance of a representation is not a reliable indicator of its REPA performance. We next show (§3) that spatial structure supplies precisely the predictive signal that global accuracy lacks.

- **Finding 1.** *Stronger global semantic information (e.g., ImageNet-1K accuracy) does not imply a better representation for generation, and can even be detrimental.*

3 SPATIAL STRUCTURE MATTERS MORE

We now formalize and test the hypothesis that spatial structure, rather than global information, is the property of a target representation that best predicts its generation performance. To this end, we introduce several straightforward metrics that quantify the spatial self-similarity structure (Shechtman & Irani, 2007) of a target representation, and we establish two results with them: every spatial structure metric correlates far more strongly with gFID than ImageNet-1K accuracy does, and, in turn, these metrics coherently explain the observations left unaccounted for in §2.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and a vision encoder $\mathcal{E}$, we build the spatial structure metrics from two ingredients: an appearance self-similarity kernel defined over patch tokens, and a spatial distance defined over their lattice positions.

- *Self-similarity.* Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ be the extracted patch representations, with $T = H \times W$ patches. Let kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ measure appearance self-similarity (Shechtman & Irani, 2007) between patch tokens. Here, we use the cosine kernel $K_{\mathcal{X}}(t, t') = \frac{(\mathbf{x}_t, \mathbf{x}_{t'})}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.
- *Spatial distance.* Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ be the spatial location of each of the T tokens in coordinate space and $d(\cdot, \cdot) \in \mathbb{N}$ be the Manhattan distance between pairs of tokens.

The *spatial* self-similarity metric then quantifies how the appearance self-similarity $K_{\mathcal{X}}$ varies as a function of the lattice distance d between patch tokens. Intuitively, larger values denote stronger spatial organization: nearby patches are more similar to one another than to distant patches. As our default instantiation, we adopt a simple correlogram contrast between local and distant pairs (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t, t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t, t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is obtained as the expectation over the patch representation $\mathcal{X}$. We fix $r_{\text{near}} = r_{\text{far}} = H/2$ throughout; the resulting correlations are robust to the precise choice of these radii. Appendix B gives exact definitions together with three alternative spatial metrics (CDS, SRSS, RMSC), each of which is similarly predictive of generation quality (see Fig. 4), indicating that our conclusions do not hinge on any single formulation.

Spatial structure correlates far more strongly with generation performance than linear probing. Our large-scale correlation analysis across 27 diverse vision encoders is clear-cut (Figure 4): the conventionally used linear probing accuracy is only weakly predictive of generation performance (Pearson $|r| = 0.26$), whereas every SSM metric attains a substantially higher correlation (Pearson $|r| > 0.85$). Across this panel of encoders, spatial structure therefore captures the property that best tracks a target representation’s generative utility.

The relationship holds across model scales. This pattern is not an artifact of a single model size. Evaluating the same relationship across three model scales (SiT-B, SiT-L, SiT-XL) in Figure 5, linear probing remains weakly correlated throughout ($|r| < 0.306$), while spatial structure remains strongly correlated with generation performance at every scale ($|r| > 0.826$). The predictive advantage of spatial structure is thus consistent across the entire model-scale range we study.

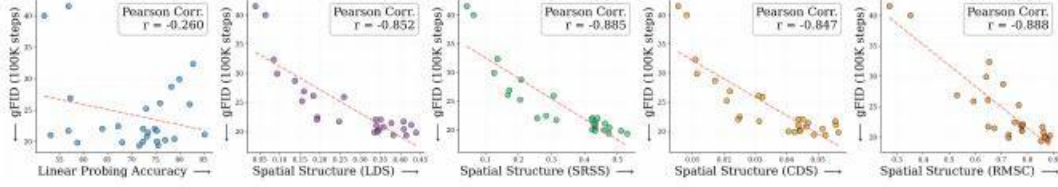


Figure 4: **Spatial structure correlates far more strongly with generation quality than linear probing.** Correlation analysis across 27 diverse vision encoders (SiT-XL/2, REPA). Linear probing accuracy is only weakly correlated with FID (Pearson $|r| = 0.260$), whereas every spatial structure metric—LDS ($|r| = 0.852$), SRSS ($|r| = 0.885$), CDS ($|r| = 0.847$), and RMSC ($|r| = 0.888$)—correlates far more strongly with generation performance. See Fig. 10 for detailed plots with encoder labels.

- **Finding 2.** *Spatial structure correlates far more strongly with generation performance than linear probing, across encoders and model scales. We next use these spatial metrics to explain trends that the accuracy-centric view leaves unexplained.*

Spatial structure explains the previously unexplained trends. As shown in §2, global performance (ImageNet-1K accuracy) fails as a predictor of representation-alignment effectiveness. We now show that the spatial metrics above resolve each of the four motivating observations, reorganizing a set of apparent anomalies into a single coherent account.

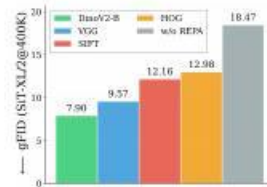
(1) *PE-Spatial-B vs. PE-Core-G:* Figure 2 shows that PE-Core-G attains far higher ImageNet-1K accuracy (82.8% vs. 53.1%) yet performs worse as a REPA target (FID 32.3 vs. 22.0). Spatial structure resolves the apparent paradox directly: despite its lower global performance, PE-Spatial-B exhibits markedly stronger spatial structure than PE-Core-G, which correctly anticipates its superior generation.

(2) *SAM2 outperforms “better” vision encoders:* as reported in §2, SAM2 attains far lower ImageNet-1K accuracy (24.1%) yet outgenerates encoders with $\sim 60\%$ higher accuracy. Figure 3a shows that this advantage is well accounted for by SAM2’s superior spatial structure.

(3) *Larger models within an encoder family underperform:* as Fig. 3b makes clear, scaling up within a family raises ImageNet-1K accuracy but can degrade spatial structure, and it is the latter that tracks the observed drop in REPA generation quality. This account also subsumes the observation of Yu et al. (2024) that larger models can yield similar or worse generation.

(4) *Injecting global information via the CLS token hurts generation:* in Figure 3c, mixing the CLS token into patch tokens improves global performance but simultaneously erodes the spatial contrast among patch tokens, causing tokens in one semantic region (e.g., a foreground object) to become spuriously similar to unrelated tokens (e.g., the background). This loss of spatial structure closely tracks the accompanying degradation in generation.

If spatial structure matters more, can classical SIFT or HOG features serve as REPA targets? They can. Although they are not the strongest targets, classical spatial descriptors—SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), and intermediate VGG features (Simonyan & Zisserman, 2014)—each yield measurable gains as REPA targets. This is direct evidence that representation alignment can benefit from spatial structure alone, with essentially no accompanying global semantic information.



Do the spatial metrics also explain the gains from REPA itself? Yes. As shown, the proposed spatial metrics account not only for the gains obtained with standard REPA but also for those of the improved training recipe, iREPA, which we introduce next in §4—indicating that a single spatial account applies both to the choice of target representation and to the gains from alignment itself.



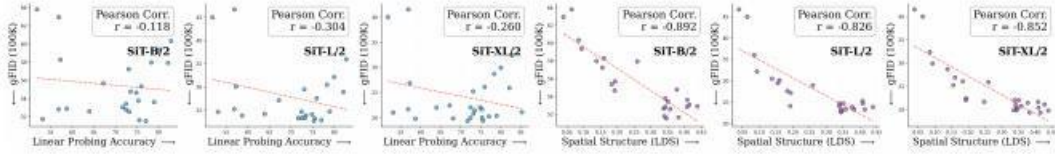


Figure 5: **Correlation analysis across model scales.** At every model scale, spatial structure (right) is far more strongly correlated with gFID than linear probing accuracy (left), showing that the predictive advantage of spatial structure is consistent across scales.

4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

Having established that spatial structure is the property that best predicts alignment quality, we now act on this principle: we introduce two minimal modifications to the original REPA training recipe, each designed to strengthen the transfer of spatial features from the teacher (the vision encoder) to the student (the diffusion transformer). Both modifications follow directly from the preceding analysis, add negligible parameters and compute, and together require fewer than four lines of code.

Convolutional projection layer instead of MLP. Standard REPA employs a three-layer MLP to project the diffusion feature dimensionality onto that of the external representation. Because this MLP acts on each token independently, it is invariant to the spatial arrangement of the token grid; as Figure 6a shows, the projection is lossy and attenuates the spatial contrast between patch tokens. We therefore replace it with a single lightweight convolutional layer (kernel size 3, padding 1) applied directly over the spatial grid. Its locality-preserving inductive bias retains the local relationships among neighboring tokens that the token-wise MLP discards, so that the spatial signal survives the projection and reaches the alignment objective intact.

Spatial normalization layer. Consistent with Siméoni et al. (2025), we find that the patch tokens of pretrained vision encoders carry a substantial global component, evidenced by the high linear probing accuracy attained by the mean of patch tokens alone (Figure 14). Figure 6b further shows that although this global component—the mean of patch tokens—inflates global performance, it does so at the expense of spatial contrast: individual patch tokens become less distinguishable, so that a token in one semantic region (e.g., a foreground object) attains high similarity to unrelated tokens (e.g., the background).

Motivated by the analysis of §3, we hypothesize that trading away this global component in favor of sharper spatial contrast strengthens the pairwise spatial signal among patch tokens and thereby improves REPA performance. We operationalize this hypothesis by applying a simple spatial normalization layer (Ulyanov et al., 2016) to the patch tokens of the target representation:

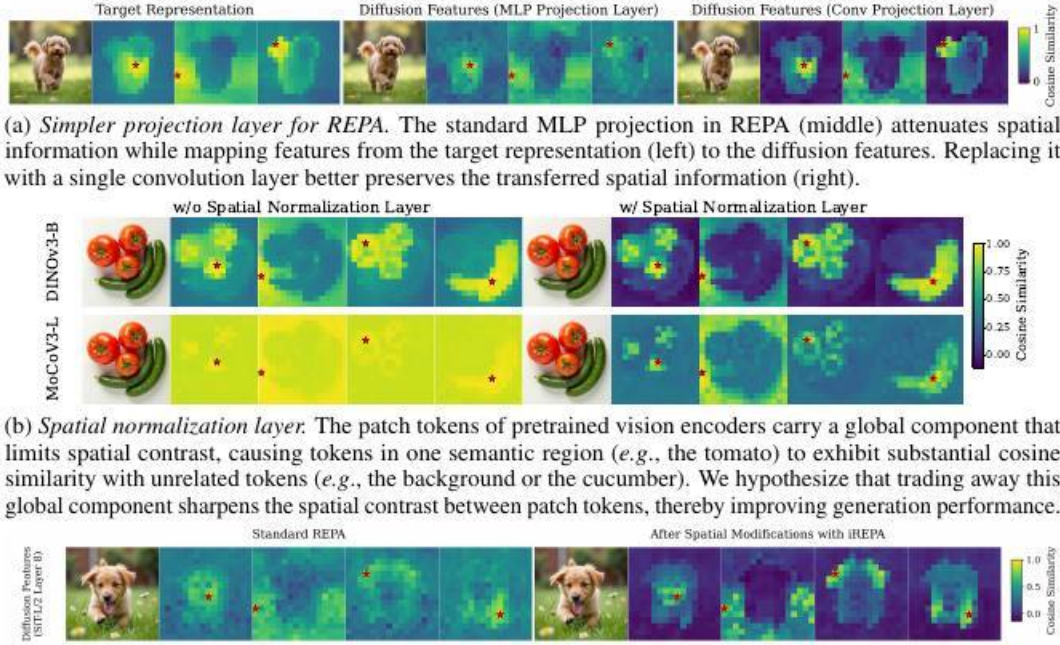
$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ denotes the patch tokens, the expectation and variance are computed across the spatial dimension, and $\epsilon = 10^{-6}$ ensures numerical stability. Subtracting the scaled spatial mean removes the dominant global component, while dividing by the spatial standard deviation rescales the residual; the layer thereby accentuates the pairwise contrast between patch tokens in the manner that the spatial-structure analysis prescribes. The scaling factor γ interpolates between leaving the global component intact ($\gamma = 0$) and removing it entirely ($\gamma = 1$), controlling how much global information is traded for spatial contrast.

Algorithm 1 Summary of key iREPA modifications to standard REPA training

```
# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
```



(c) Overall impact of the improved training recipe (iREPA) on the spatial structure of the diffusion features. Figure 6: Motivating the two minimal modifications (iREPA) that enhance the transfer of spatial features from the target representation to the diffusion features. All results reported with SiT-L/2 and REPA at 100K.

4.1 EXPERIMENTS

We validate the improved training recipe through an extensive suite of experiments, centered on ImageNet 256×256 and extended to higher resolution, pixel-space diffusion, and text-to-image generation in the appendix. The evaluation is organized around three research questions that together probe the generality of the approach:

- Does iREPA consistently accelerate the convergence of diffusion transformers over REPA across diverse external representations? (Figure 7, 8, 12, 13, 15)
- Does iREPA scale with model size and generalize across variations in the training setup? (Table 1 [a,b,c], 2, 4, 5, 6, and Figure 12, 16)
- Does iREPA transfer to more recent representation-alignment methods such as REPA-E (Leng et al., 2025a) and MeanFlow with REPA (Geng et al., 2025), and to pixel-space diffusion models such as JiT with REPA (Li & He, 2025b)? (Table 3 [a,b])

Convergence Speed. We evaluate convergence across a diverse panel of vision encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and two model sizes (SiT-XL/2 and SiT-B/2), with results in Fig. 7. For every combination of target representation and model size, iREPA converges faster than baseline REPA, indicating that the benefit of accentuating spatial transfer is systematic across these settings rather than specific to a particular encoder or scale.

Target representation. We assess generalization across vision encoders in Fig. 8 and Table 4. iREPA improves generation quality for every encoder tested, without exception, and this uniformity extends to the full panel of 27 encoders; comprehensive comparisons are reported in Appendix C, E.

Encoder size. We next vary encoder capacity within the PE family—PE-B (90M), PE-L (320M), and PE-G (1.88B)—in Table 1a, using SiT-XL/2 at 100K iterations with vanilla REPA as the baseline. iREPA improves generation quality (FID) at every encoder size, and, notably, its relative gain grows with encoder capacity (22.2% for PE-B, 38.8% for PE-L, 39.6% for PE-G). This is a particularly favorable property: the larger encoders whose spatial structure baseline REPA fails to exploit are precisely those from which iREPA recovers the most.

Enc. Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
PE-B (90M)	59.6	22.5	6.23	0.63	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60
+iREPA	76.3	17.6	6.36	0.65	0.61
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59
+iREPA	70.8	19.5	6.35	0.64	0.61

(a) Vision Encoder size

Model Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
SiT-B	27.5	49.50	7.00	0.46	0.59
+iREPA	34.1	43.37	7.87	0.50	0.60
SiT-L	55.7	24.10	6.25	0.62	0.60
+iREPA	66.9	20.28	6.14	0.63	0.62
SiT-XL	70.3	19.06	5.83	0.65	0.61
+iREPA	77.9	16.96	6.26	0.66	0.61

(b) Model size

Aln. Depth	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	41.7	36.0	6.61	0.52	0.62
Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	42.4	35.3	6.90	0.53	0.62
Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	38.3	38.9	7.19	0.51	0.62

(c) Encoder depth (SiT-B/2)

Table 1: **Variation in training settings.** iREPA delivers consistent FID gains over baseline REPA along all three axes of variation—encoder size (a), model size (b), and alignment depth (c)—with relative gains that grow with both encoder and model size. Unless otherwise specified, results use DINOv2 as the encoder, SiT-XL/2, 100K steps, and vanilla REPA as the baseline; the iREPA rows are highlighted, and per-metric bests are in bold.

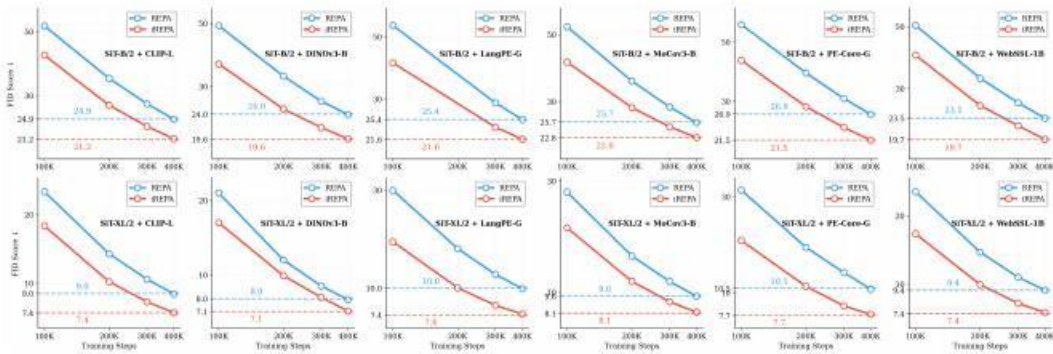


Figure 7: **Accentuating spatial features consistently improves convergence speed.** FID convergence curves for REPA and iREPA across a panel of vision encoders and two model sizes (SiT-XL/2 and SiT-B/2); iREPA converges faster than baseline REPA in every case.

Scalability. We examine behavior across model scales in Table 1b (SiT-B, SiT-L, SiT-XL). The spatial improvements not only hold at every scale but yield larger relative FID gains for larger models, indicating that iREPA scales favorably with model size rather than saturating as capacity grows.

Encoder depth. Table 1c varies the alignment depth—the layer at which the alignment loss is applied—using SiT-B/2 at 100K iterations with DINOv3-B as the target representation. iREPA improves over baseline REPA at every alignment depth, indicating that the benefit is largely insensitive to the specific layer at which alignment is applied.

Ablation on the individual components. Table 2 isolates the contribution of each component across four encoders (DINOv2-B, DINOv3-B, WebSSL-1B, PE-Core-G) at 100K steps with SiT-XL/2. Each component—the spatial normalization layer and the convolutional projection layer—independently improves generation quality (FID) over baseline REPA, and their combination is strictly best for every encoder. This pattern is consistent with the two modifications addressing complementary sources of spatial-information loss: one at the target representation and one at the projection.

Training recipe. Finally, we test whether the benefit transfers beyond standard REPA to more recent alignment recipes, namely REPA-E (Leng et al., 2025a) and MeanFlow with REPA (Geng et al., 2025). Table 3 shows that iREPA’s spatial improvements yield convergence gains under both recipes, indicating that accentuating spatial transfer generalizes across alignment recipes rather than being a tweak specific to the original REPA formulation.

Classifier-free guidance and longer training. Table 4 reports generation quality on SiT-XL/2, both with and without classifier-free guidance, at 100K and 400K training iterations. iREPA improves over baseline REPA for every encoder across all four combinations of guidance setting and training budget; the gains therefore hold under classifier-free guidance and persist at the longer 400K-iteration budget rather than being confined to the early, fixed-step regime.

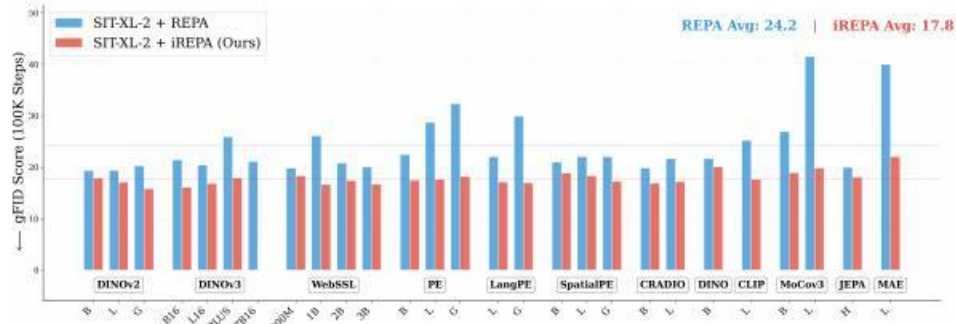


Figure 8: **Variation in target representation.** Across all 27 vision encoders, accentuating the transfer of spatial features from the target representation to the diffusion features (iREPA) consistently improves convergence speed, without a single exception. See Appendix C for additional results across diverse settings.

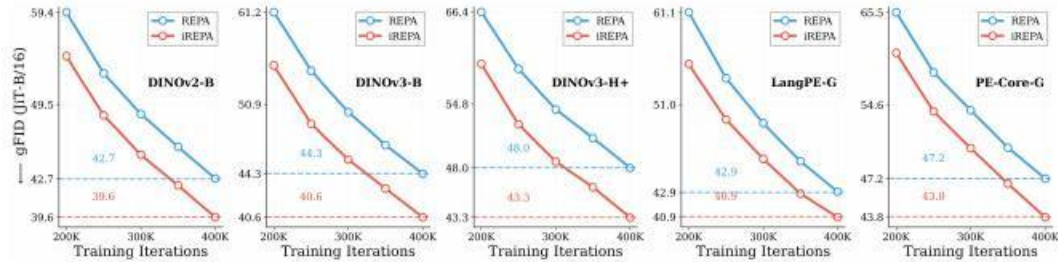


Figure 9: **Convergence comparison with pixel-space diffusion (JiT).** FID convergence curves comparing REPA and iREPA within pixel-space JiT (Li & He, 2025b) across several vision encoders. Accentuating the transfer of spatial structure consistently improves convergence speed for every encoder, showing that the principle carries over from latent-space to pixel-space diffusion. All results use JiT-B/16 (Li & He, 2025b), batch size 256, and no classifier-free guidance. See Table 10 for further results.

Pixel-space diffusion (JiT). We further evaluate iREPA on pixel-space diffusion models, using JiT-B (Li & He, 2025b). Figure 9 compares REPA and iREPA within JiT (Li & He, 2025b): accentuating spatial-information transfer again yields consistently faster convergence across vision encoders (e.g., DINOv2, DINOv3, PE), so the principle carries over from latent-space to pixel-space diffusion.

- **Finding 3.** Accentuating the transfer of spatial structure from the target representation to the diffusion features consistently boosts the convergence speed of representation alignment (REPA), across encoders, model sizes, alignment depths, and training recipes.

5 RELATED WORK

We situate our contribution among the most directly relevant lines of work here, deferring an extended discussion to Appendix H.

Representation alignment for generation. A growing body of work leverages external representations to improve diffusion model training (Pernias et al., 2023; Fuest et al., 2024). In particular, recent methods (Yu et al., 2024; Yao & Wang, 2025; Leng et al., 2025a;b; Kouzelis et al., 2025) show that substantial gains follow from aligning internal diffusion features with the clean image features of a pretrained vision encoder, and the idea has since been extended to video (Zhang et al., 2025) and 3D generation (Wu et al., 2025) as well as to the training of unified models (Ma et al., 2025a). Despite these consistent empirical successes, the property of the target representation responsible for the acceleration has remained unresolved. We address this gap by identifying *which* aspect of

Method	DINOv2-B			DINOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation on the individual components.** Across all four encoders (DINOv2-B, DINOv3-B, WebSSL-1B, PE-Core-G), the spatial normalization layer and the convolutional projection layer (§4) each yield substantial FID gains over baseline REPA, and their combination (highlighted) is strictly best for every encoder—consistent with the two modifications addressing complementary sources of spatial-information loss. All results are reported at 100K steps with SiT-XL/2; per-metric bests are in bold.

Encoder	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	w/o CFG				w/ CFG			
						4 NFE		1 NFE		4 NFE		1 NFE	
						IS↑	FID↓	IS↑	FID↓	IS↑	FID↓	IS↑	FID↓
WebSSL-1B	52.8	26.5	5.20	0.620	0.585								
+iREPA-E	87.0	13.2	5.28	0.699	0.598								
PE-G	50.9	25.9	5.68	0.612	0.576								
+iREPA-E	80.0	16.4	5.40	0.667	0.616								
DINOv3-B	82.2	14.4	4.68	0.694	0.596								
+iREPA-E	93.6	11.7	4.57	0.703	0.613								
DINOv2-B	87.5	12.9	5.40	0.708	0.586								
+iREPA-E	91.3	12.1	4.86	0.712	0.602								
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74					
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69					
DINOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55					
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32					
DINOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17					
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62					

(a) REPA-E (SiT-XL/2)

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Variation in training recipes.** Layering our spatial improvements onto REPA-E (a, SiT-XL/2) and MeanFlow with REPA (b, SiT-B/2) yields consistent gains under both recipes, indicating that the benefit generalizes across alignment recipes. For MeanFlow we report results at 1 and 4 NFEs, using a CFG value of 2.0. All results are reported at 100K training iterations; the iREPA rows are highlighted, and per-metric bests are in bold.

the target representation—its spatial structure—best predicts generation quality, and we convert that understanding into a concrete, improved training recipe.

The spatial–global information trade-off in pretrained vision encoders. A parallel line of work examines the trade-off between global and spatial information in pretrained vision encoders (Bolya et al., 2025; Siméoni et al., 2025). Siméoni et al. (2025) show that continued self-supervised training can raise the similarity between the global CLS token and the patch tokens, degrading performance on dense spatial tasks, whereas Bolya et al. (2025); Heinrich et al. (2024) deliberately train spatially tuned models for such tasks. Our results connect this trade-off directly to generation: for the representation-alignment settings we study, the spatial structure of a vision encoder is a stronger determinant of generation quality than its global information. This connection suggests a reconsideration of how external representations for generation are selected and trained.

6 CONCLUSION

This paper addresses a foundational question about representation alignment: whether its effectiveness derives from the global information or the spatial structure of the target representation. Through a large-scale empirical analysis across 27 vision encoders and three model scales, we establish that spatial structure—not global information—is the property that best predicts alignment quality in the diffusion-transformer settings we study, and we introduce a training-free spatial-structure metric that anticipates a representation’s generative utility far more reliably than linear probing accuracy. Acting on this account, we propose two minimal modifications that accentuate the transfer of spatial information from the target representation to the diffusion features. The resulting method, iREPA, consistently accelerates the convergence of REPA across vision encoders, model sizes, alignment depths, and training recipes, with gains that persist under classifier-free guidance and at longer training budgets, at negligible added cost. The analysis is primarily empirical and correlational, and we regard a fuller causal characterization—and a broader test beyond ImageNet-style generation—as natural next steps. Even so, our findings reframe how representation alignment is understood and

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+iREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DINOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+iREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DINOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+iREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DINOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+iREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+iREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+iREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+iREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+iREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **Results across encoders, with and without classifier-free guidance.** Using SiT-XL/2 as the base model, iREPA improves generation quality over baseline REPA for every encoder, in both the guided and unguided settings and at both 100K and 400K training iterations—so the gains hold under classifier-free guidance and persist at the longer budget. See Appendix E for detailed results on SiT-B/2 and SiT-L/2 across all encoders.

chart a principled path toward selecting, measuring, and leveraging external representations for the more efficient training of generative models.

REPRODUCIBILITY STATEMENT

We report all implementation details and hyperparameters—including the exact form of the convolutional projection and spatial normalization layers, the spatial-structure metrics, and the training and evaluation protocols—in Appendix G, so that the method can be reimplemented directly from the paper. Our code, model checkpoints, and analysis results will be released upon acceptance.

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A DETAILED CORRELATION ANALYSIS

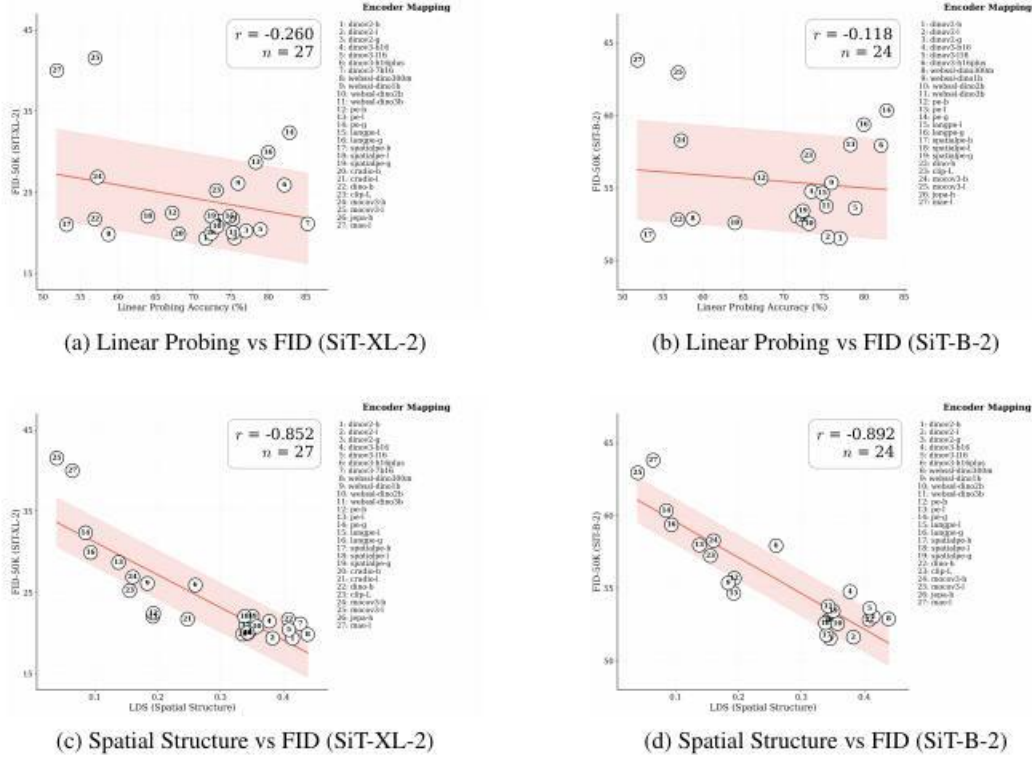


Figure 10: Detailed correlation analysis with encoder labels. The top row plots linear probing accuracy against FID for the SiT-XL-2 and SiT-B-2 models, and the bottom row plots spatial structure (LDS) against FID; each point is annotated with its encoder number (see legend). At both model scales, the spatial structure metric correlates substantially more strongly with generation quality than linear probing accuracy does.

B SPATIAL SELF-SIMILARITY METRICS

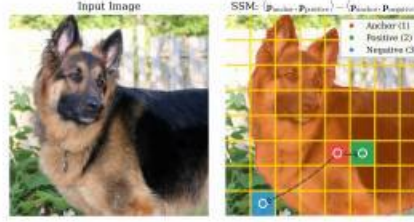


Figure 11: **Explaining the semantic-region self-similarity (SRSS) metric.** Visual explanation of the SRSS metric: for each image, we compute a mask using SAM2 and select anchor–positive–negative triplets. The metric measures the difference in cosine similarity between anchor–positive pairs (within the mask) and anchor–negative pairs (outside it). Larger values indicate that patches within the same semantic region are more similar to one another than to patches in unrelated regions, signaling better preservation of spatial structure.

Setup. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we extract patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ with $T = H \times W$ and indices $t \in [T]$ placed on an $H \times W$ lattice. Let $d : [T] \times [T] \rightarrow \mathbb{N}$ be the Manhattan distance, and let the (cosine) self-similarity kernel be

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

a standard local self-similarity measure (Shechtman & Irani, 2007).

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ where $T = H \times W$ is the spatial grid of patches (patch index set $[T]$). Also let $d : [T] \times [T] \rightarrow \mathbb{N}$ be Manhattan distance on the $H \times W$ grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ measuring self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ measures how self-similarity K between patch tokens varies with lattice distance d . Intuitively, larger values indicate stronger spatial organization (near patches more similar to each other than far away patches).

We now define several straightforward metrics that quantify the spatial self-similarity structure of a target representation. Each is a functional of the self-similarity kernel $K_{\mathcal{X}}$ and the lattice distance d , and each captures a distinct facet of spatial organization, so that our conclusions do not rest on any single definition.

Local vs. Distant Similarity (LDS). We first consider a simple correlogram contrast between local and distant pairs (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}],$$

where r_{near} and r_{far} are hyperparameters, set by default to $r_{\text{near}} = r_{\text{far}} = H/2$; the induced correlations are robust to their exact values. A larger LDS indicates that, *on average*, spatially proximate patches are more similar than distant ones—the signature of preserved spatial organization.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ over distances $\delta \in \{1, \dots, \Delta\}$, fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$ to it, and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where $\hat{\beta}$ is the fitted slope. A larger CDS reflects faster similarity decay with distance, and hence stronger spatial organization.

Semantic-Region Self-Similarity (SRSS). For each image, we sample a binary mask $\mathbf{M} \in \mathbb{R}^{H \times W}$ enclosing a random object using SAM2 (Ravi et al., 2024), downsampled to $H \times W$, and draw anchor, positive, and negative points: the anchor and positive are sampled within the mask (semantically related), and the negative outside it (less related to the anchor). A representation that preserves spatial structure should render the anchor–positive pair more similar than the anchor–negative pair;

we therefore define the Spatial Structure Metric (SSM) as the expected gap between these two similarities:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

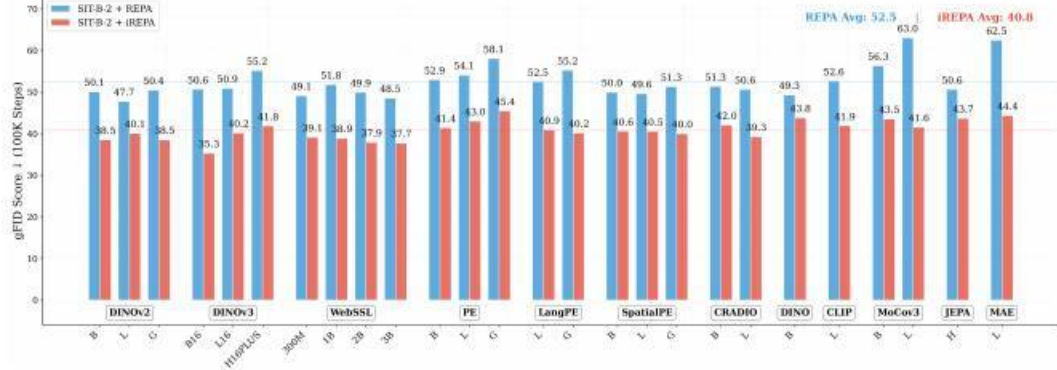
where positive points lie within Manhattan distance $d \leq r_{\text{near}}$ of the anchor inside the mask, and negative points lie at distance $d \geq r_{\text{far}}$ outside it.

RMS Spatial Contrast (RMSC). Finally, we consider a simple metric that captures the dispersion of patch-token representations. Given ℓ_2 -normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

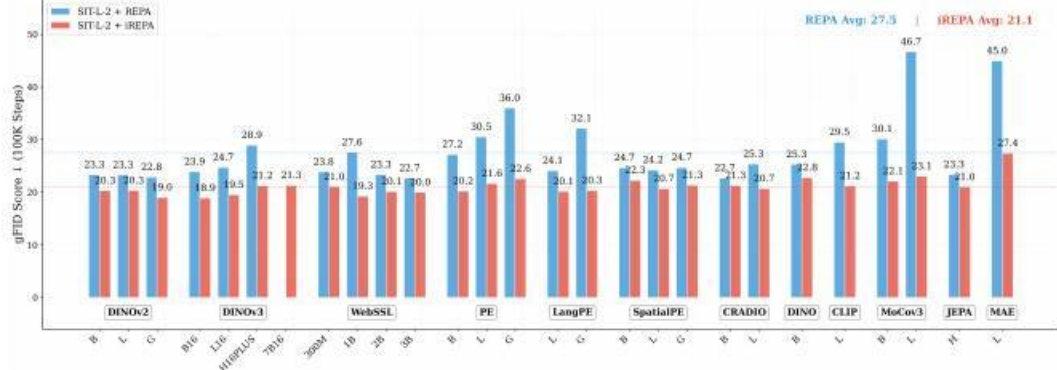
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ is the mean of the normalized features over all spatial locations. A higher RMSC indicates greater spatial diversity among features and thus preserved spatial structure, whereas a lower value signals a more uniform, spatially collapsed representation—the regime that arises when a dominant global component erodes spatial structure (Siméoni et al., 2025).

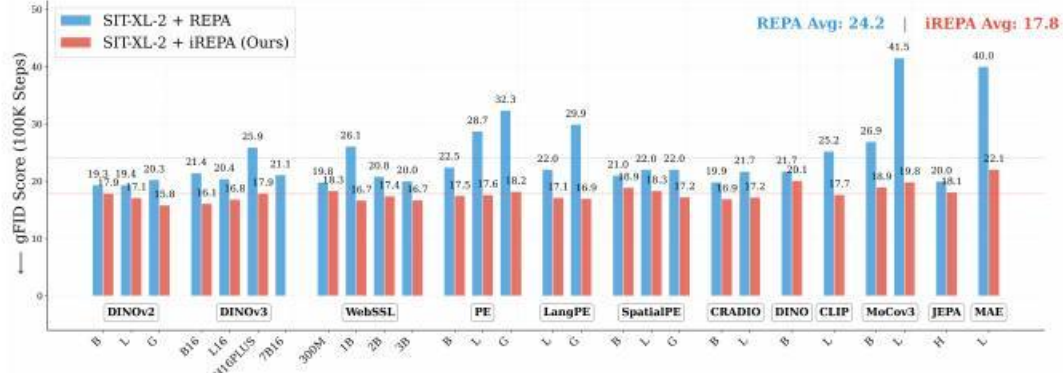
C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



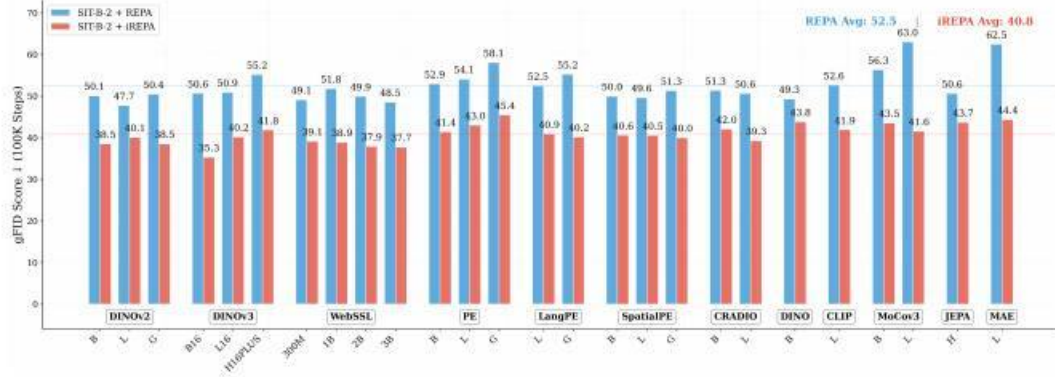
(b) SiT-L-2 (458M parameters)



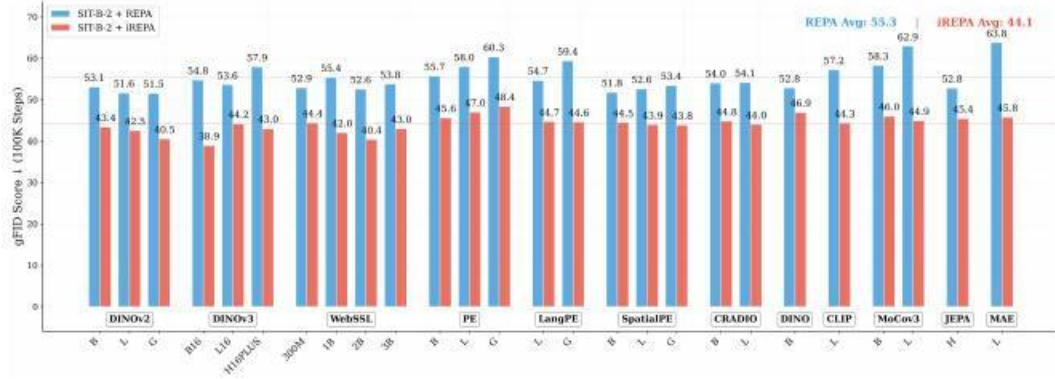
(c) SiT-XL-2 (675M parameters)

Figure 12: **Variation in vision encoders.** Ablations comparing REPA and iREPA across a range of vision encoders for the SiT-B/2, SiT-L/2, and SiT-XL/2 models. iREPA improves generation quality for every encoder at every model size.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Effect of encoder depth on iREPA performance.** SiT-B-2 performance under two encoder depths (6 vs. 8 layers) across a range of vision encoders. iREPA improves generation quality at both depths—with 8 layers slightly ahead—and both configurations exhibit substantial FID gains once the spatial modifications are applied, indicating that the benefit is robust to encoder depth.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: **SiT-B/2 performance across vision encoders at 400K iterations.** iREPA improves generation quality over the vanilla-REPA (Yu et al., 2024) baseline for every encoder, with the highlighted iREPA rows attaining the better FID (and, for nearly all encoders, the better IS, precision, and recall). All baselines are trained with vanilla REPA.

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: **SiT-L/2 performance across vision encoders at 400K iterations.** iREPA improves generation quality for every encoder, with the largest gains in FID and IS. The highlighted iREPA rows mark the improved model, and per-metric bests are in bold. All baselines are trained with vanilla REPA (Yu et al., 2024).

F ADDITIONAL EXPERIMENTAL RESULTS

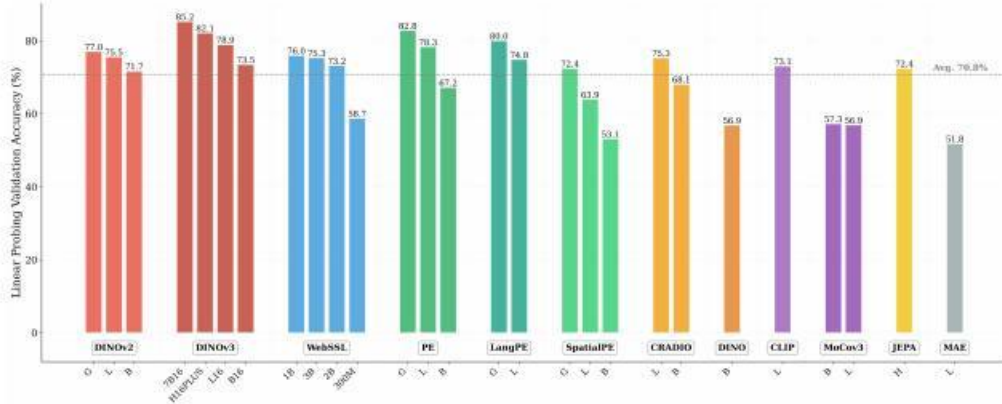


Figure 14: **Global information in the mean of patch tokens.** Beyond the [CLS] token, the mean of patch tokens itself carries substantial global semantic information. While this raises global performance, it reduces the contrast between individual patch tokens and thereby hinders spatial-structure transfer. Removing part of this global component—by subtracting the scaled mean of patch tokens—improves spatial-structure transfer during representation alignment (§4).

Vision Encoder	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Generation quality with classifier-free guidance.** REPA vs. iREPA with CFG (scale 2.0) across vision encoders, using SiT-XL/2 trained for 400K iterations with 250 sampling steps. iREPA improves both IS and FID for every encoder, so the gains persist under classifier-free guidance and at the longer 400K-iteration budget. The iREPA rows are highlighted, and per-metric bests are in bold.

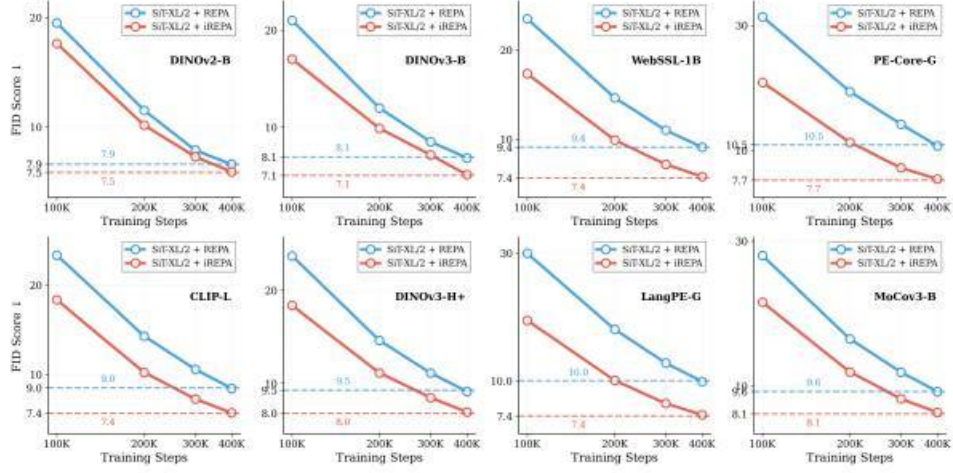


Figure 15: **Accentuating spatial features consistently improves convergence speed.** FID convergence curves for SiT-XL/2 under REPA and iREPA; iREPA reaches a given FID in fewer iterations across the encoders shown.

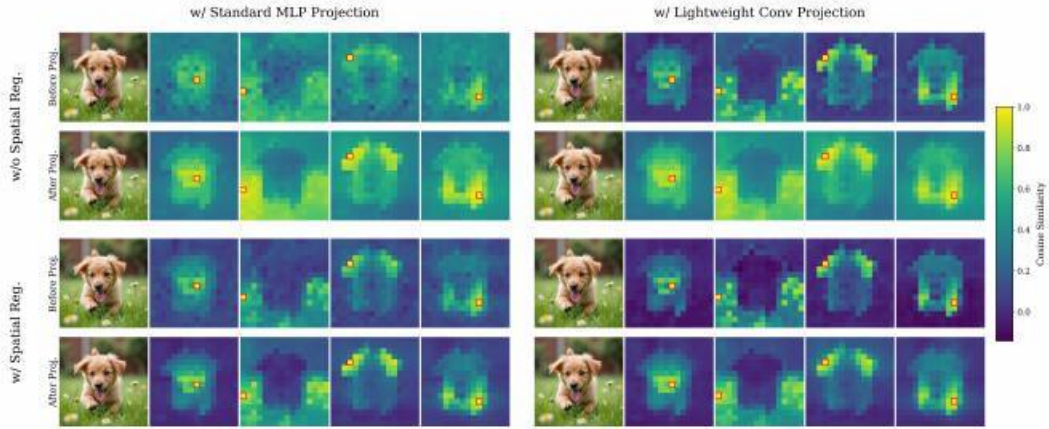


Figure 16: **Visualizing the impact of the two minimal modifications that enhance spatial feature transfer.** First, the standard MLP projection (top-left) attenuates spatial information while mapping features from the pretrained encoder (after projection) to the diffusion model (before projection); a single convolution layer better preserves the transferred spatial information (top-right). Second, vision-encoder features often exhibit limited spatial contrast (§2), causing tokens in one semantic region (e.g., the dog) to exhibit substantial cosine similarity with unrelated tokens (e.g., the background). We address this with a simple spatial normalization layer that accentuates the spatial contrast in the target representation, thereby improving generation performance. Combining both modifications is best (see Table 2).

G IMPLEMENTATION DETAILS

Training setup. Unless otherwise specified, we follow the setup of REPA (Yu et al., 2024). All training uses the ImageNet (Deng et al., 2009) training split. For preprocessing we adopt the ADM (Dhariwal & Nichol, 2021) protocol, center-cropping and resizing each image to 256×256 resolution, and we use the `stabilityai/sd-vae-ft-mse` VAE (AI, n.d.) throughout diffusion training and inference. The spatial normalization layer uses $\gamma \in [0.6, 0.8]$, and the projection layer is a single convolutional layer with kernel size 3 and padding 1, which replaces REPA’s three-layer MLP; both modifications add negligible parameters and leave per-step training cost essentially unchanged relative to vanilla REPA. Optimization uses AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with a constant learning rate of 1×10^{-4} and a global batch size of 256. For efficient and stable training we employ `bfloat16` mixed precision, `torch.compile`, gradient clipping, and an exponential moving average (EMA) of the generative model.

For the REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a) experiments we use the official open-source implementations. For MeanFlow (Geng et al., 2025), we obtained the implementation (including REPA) from the original authors and adapted it to incorporate the two iREPA modifications (§4).

Evaluation. For image generation, we follow the ADM evaluation setup (Dhariwal & Nichol, 2021) exactly, reporting generation quality via Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), and precision (Prec.) and recall (Rec.) (Kynkäänniemi et al., 2019), each computed over 50K generated images. Following REPA (Yu et al., 2024), sampling uses the SDE Euler–Maruyama sampler with 250 steps; for JiT (Li & He, 2025a) we use 50 inference steps, matching the official implementation.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Spatial feature extraction has long been foundational to computer vision, with classical descriptors such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) providing robust local descriptors for tasks ranging from object detection to image matching. These handcrafted features excel at capturing local spatial patterns and geometric invariances, yet recent generative modeling has drawn almost exclusively on modern self-supervised representations with strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our findings motivate a complementary perspective: because spatial structure is what best tracks generation quality, even classical spatial descriptors can improve diffusion training when properly aligned. This points to a broader opportunity to draw on the full spectrum of spatial feature extractors, from handcrafted descriptors to learned representations, provided they preserve strong spatial coherence.

Pretrained visual encoders for generative models. Pretrained visual encoders have supported generative models in several roles: as discriminators that accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers in adversarial distillation for diffusion models (Sauer et al., 2024; Kang et al., 2024), and, more recently, as alignment targets. For GANs (Goodfellow et al., 2020), pretrained features have both improved convergence speed and enabled scaling to large datasets, as demonstrated by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) using CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation uses pretrained encoders to steer student networks toward higher-fidelity samples, with clear perceptual-quality gains. REPA (Yu et al., 2024), in particular, aligns diffusion features with external encoders and shows that representation alignment improves both convergence and generation quality. We build on this direction but reorient its emphasis: what governs alignment quality, in the settings we study, is the preservation of spatial structure rather than the discriminative capability of the encoder.

Denosing transformers. Transformers have become the dominant backbone for scalable generative modeling, spanning diffusion transformers (Peebles & Xie, 2023) and flow-matching variants (Ma et al., 2024). Recent architectures push this further: GenTron (Chen et al., 2023) scales transformers beyond 3B parameters for text-to-image synthesis; U-ViT (Bao et al., 2023) shows that pure transformer backbones without U-Net inductive biases remain competitive; and ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore autoregressive formulations that couple denoising

with sequential generation to trade off quality against speed. Across these advances, training from scratch remains computationally expensive, frequently demanding millions of iterations to reach high generation quality. Representation alignment methods such as REPA (Yu et al., 2024) sharply accelerate this convergence, but the property responsible for the acceleration had remained unclear. Our analysis identifies that property: the benefit tracks the preservation of spatial structure rather than semantic alignment, which helps explain why some encoders accelerate training far more than others and offers a principle for designing more effective alignment strategies. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denoising as a self-supervised learning task. The link between denoising and representation learning has been examined from several angles. Early work by Abstreiter et al. (2021) adapted diffusion objectives for representation learning, showing that denoising naturally yields meaningful features; SODA (Hudson et al., 2024) pairs a diffusion model with an information bottleneck to learn compact representations; and Chen et al. (2024) deconstruct diffusion models to show that a simple denoising autoencoder suffices for strong self-supervised performance. More recently, Wang & He (2025) introduce a dispersive loss that encourages internal representation diversity and improves generation without external encoders, and Jiang et al. (2025) propose Self-Representation Alignment (SRA), which aligns a diffusion transformer’s latent features across noise levels to provide self-guidance without an auxiliary model. Collectively, these works cast denoising as a fundamental self-supervised task that promotes robust features. Our findings refine this view: within representation alignment, it is the spatially rich structure of self-supervised encoders—rather than their global semantic content—that best accounts for the improvement in generation, identifying spatial-structure preservation as the operative factor in the synergy.

Scaling self-supervised vision encoders. Recent years have brought striking progress in scaling self-supervised vision models to unprecedented sizes and datasets. DINOv3 (Siméoni et al., 2025) trains a 7B-parameter ViT on 1.7 billion unlabeled images by aligning representations across augmentations; WebSSL (Fan et al., 2025) shows that models trained on more than 2 billion images can match language-supervised models such as CLIP (Radford et al., 2021) on vision-language tasks without any language supervision; and C-RADIO (Heinrich et al., 2024) distills multiple teachers into versatile encoders that generalize across visual domains. Complementing these, I-JEPA (Assran et al., 2023) learns by predicting masked regions in representation space rather than in pixel space, SAM (Ravi et al., 2024) specializes in promptable segmentation via large-scale supervised training, and the Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features often surpass final representations for dense prediction. These models are almost always benchmarked on global tasks such as image classification; our work highlights a gap in that practice: for representation alignment, strong discriminative performance does not translate into better generation. On the contrary, within the encoders we study, those that most benefit diffusion training are the ones that preserve spatial structure, largely irrespective of classification accuracy. This suggests that the evaluation of self-supervised encoders intended for generative applications should treat spatial coherence as at least as important as semantic understanding.

I NOTE ON LLM USAGE

We used GPT-5 (OpenAI, 2025) to explore alternative formulations of the spatial structure metrics discussed in §3. All figures were generated directly from our experiment logs and checkpoints using Claude-Code (Anthropic, 2025). We additionally used LLM assistance to search for and organize the related work in Appendix H, and Cursor for portions of the writing.

J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

Spatial structure metrics can measure the representation gap. Yu et al. (2024) quantify the representation gap through linear probing, attributing REPA’s effectiveness to the resulting increase in validation accuracy. We show that spatial structure metrics (SSM) provide an equally natural—and, given our analysis, a more apt—measure of this gap and of the improvements across models. As Figure 17 illustrates, representation alignment closes the SSM gap between the diffusion features and those of a pretrained encoder such as DINOv2, which recasts alignment as the bridging of a *spatial* feature gap.

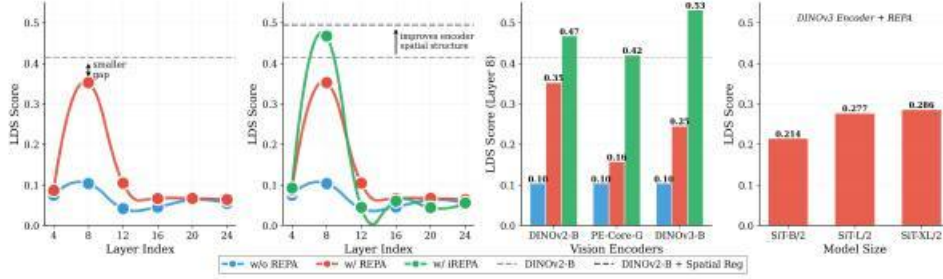


Figure 17: **Representation alignment as bridging the spatial feature gap.** a) Representation alignment can also be viewed as bridging the spatial feature gap between the diffusion features and the vision-encoder patch features. b) iREPA (§4) accentuates the spatial features of the vision encoder—at the cost of some global information—and thereby improves generation performance. c) iREPA consistently improves performance across vision encoders. d) The spatial-structure improvements scale with model size. All results are reported with SiT-XL/2 at 400K iterations.

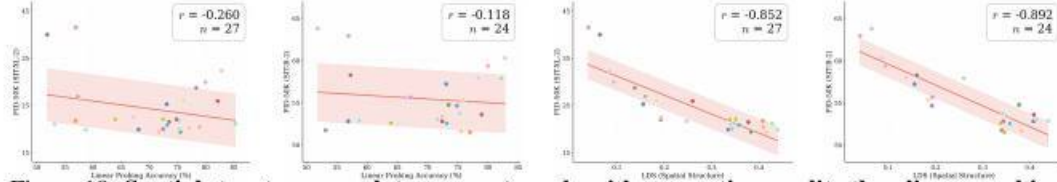


Figure 18: **Spatial structure correlates more strongly with generation quality than linear probing.** Correlation analysis across 27 vision encoders. **Left two:** linear probing accuracy vs. FID for SiT-XL-2 (Pearson $r = -0.26$) and SiT-B-2 ($r = -0.12$) is only weakly correlated. **Right two:** spatial structure (LDS) vs. FID for SiT-XL-2 ($r = -0.85$) and SiT-B-2 ($r = -0.89$) is strongly correlated. See Figure 10 in the Appendix for detailed plots with encoder labels.

K ADDITIONAL ANECDOTAL COMPARISONS

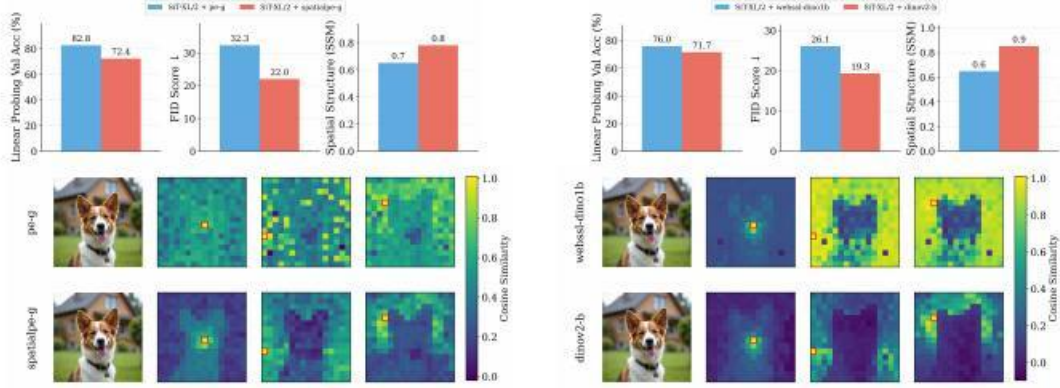


Figure 19: **Motivating examples from recent SSL encoders — spatial structure matters.** **Top row:** the metrics comparison reveals an inverse relationship between ImageNet accuracy and generation quality. **Left:** PE-g attains higher accuracy (82.8% vs. 72.4%) but worse FID (32.3 vs. 22.0) than Spatial-PE-g, together with far lower spatial structure (LDS: 0.1 vs. 0.4). **Right:** WebSSL-dino1b attains higher accuracy (76.0% vs. 71.7%) but worse FID (26.1 vs. 19.3) than DINOv2-b, together with weaker spatial structure (LDS: 0.2 vs. 0.4). **Bottom row:** the spatial cosine-similarity visualizations are consistent with this account—the encoders that generate better (Spatial-PE-g, DINOv2-b) retain clear spatial coherence, whereas those optimized for classification (PE-g, WebSSL) lose spatial structure.

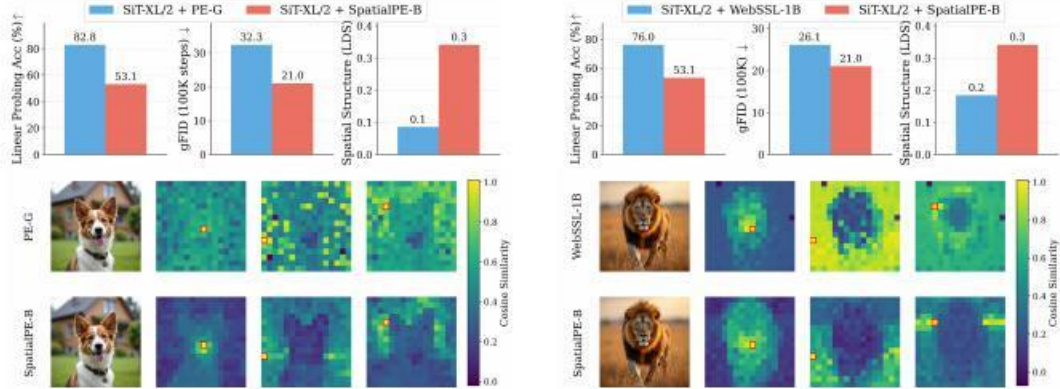


Figure 20: **Motivating examples — spatial structure matters.** **Top:** the metrics comparison shows an inverse relationship between ImageNet accuracy and generation quality. **Left:** PE_{core}-G, despite substantially higher validation accuracy (82.8% vs. 53.1%), generates worse than PE_{Spatial}-B (Bolya et al., 2025). **Right:** likewise, WebSSL-1B (Fan et al., 2025) attains far higher global performance (76.0% vs. 53.1%) yet worse generation. **Bottom:** spatial structure, by contrast, is the better predictor of generation quality. See §3 for the spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of Spatial Normalization. We visualize how spatial normalization reshapes vision-encoder representations. Pretrained encoders often carry a substantial global component, or overlay, that suppresses the contrast between spatial tokens: a token in one semantic region can be strongly correlated with tokens in unrelated regions (e.g., the background), diluting the spatial distinctiveness of the features. Spatial normalization removes this global overlay and thereby amplifies the contrast between distinct spatial tokens, so that the model preserves local spatial structure while attenuating the global information that would otherwise interfere with generation quality.

The figures below make this effect concrete across several examples, illustrating how spatial normalization reshapes the feature representations to foreground spatial structure:

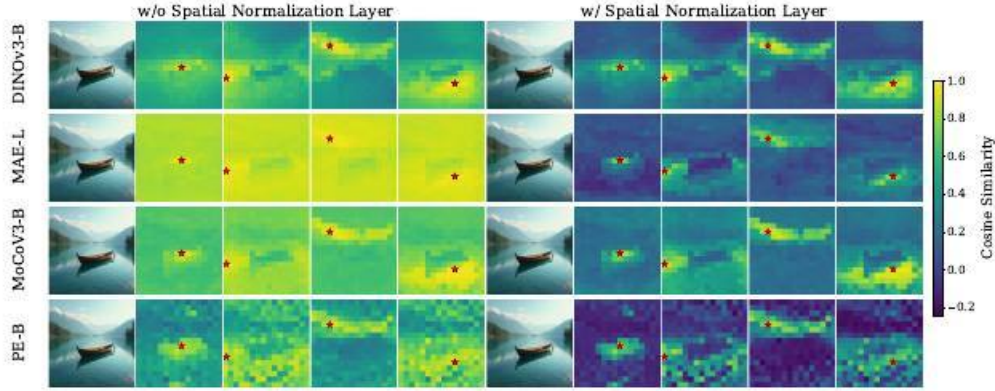


Figure 21: **Visualizing the impact of spatial normalization** (Example 1). The heatmaps show token-similarity patterns before and after spatial normalization. Without normalization (left), the global component induces high correlations across unrelated regions; with spatial normalization (right), local spatial structure is enhanced and global interference is reduced, yielding more distinct semantic boundaries.

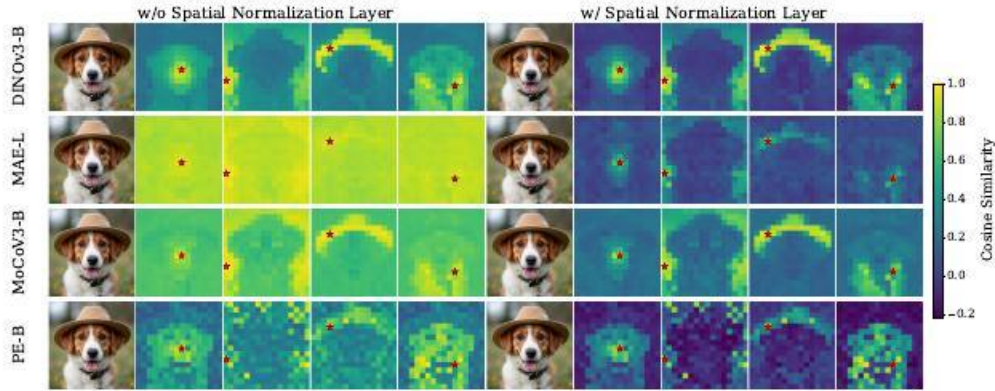


Figure 22: **Visualizing the impact of spatial normalization** (Example 3). The heatmaps show token-similarity patterns before and after spatial normalization. Without normalization (left), the global component induces high correlations across unrelated regions; with spatial normalization (right), local spatial structure is enhanced and global interference is reduced, yielding more distinct semantic boundaries.

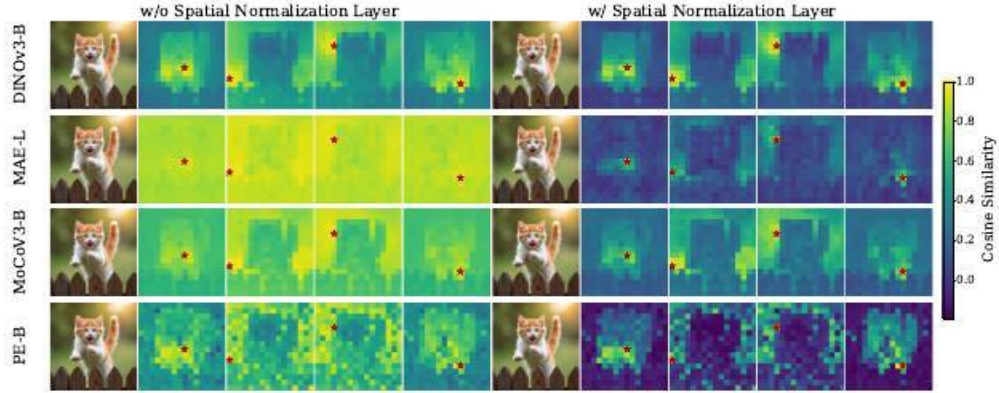


Figure 23: **Visualizing the impact of spatial normalization** (Example 2). Feature-similarity maps show how spatial normalization improves spatial contrast. The original features (left) exhibit a global overlay that blurs the distinction between foreground and background; after normalization (right), the spatial tokens become more locally coherent, with clearer separation between semantic regions.

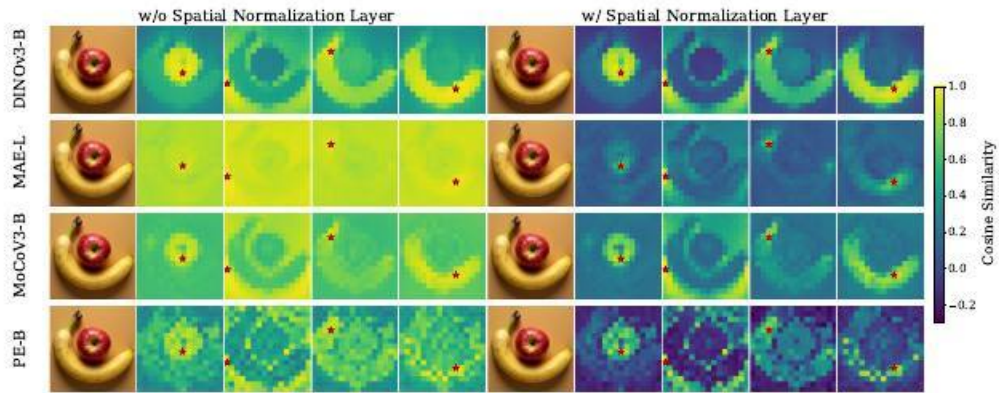


Figure 24: **Visualizing the impact of spatial normalization** (Example 4). The heatmaps show token-similarity patterns before and after spatial normalization. Without normalization (left), the global component induces high correlations across unrelated regions; with spatial normalization (right), local spatial structure is enhanced and global interference is reduced, yielding more distinct semantic boundaries.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. To assess whether the proposed spatial improvements hold at higher resolution, we conduct additional experiments on ImageNet-512 (Deng et al., 2009). Following the REPA (Yu et al., 2024) setup, we report results over a range of pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.), using SiT-XL and SiT-B with 50 NFE at inference and without classifier-free guidance. Fig. 25 compares REPA before and after the spatial improvements (iREPA). The spatial improvements carry over to the higher resolution, yielding consistent gains across every pretrained encoder considered (DINOv2, DINOv3, WebSSL, PE, etc.).

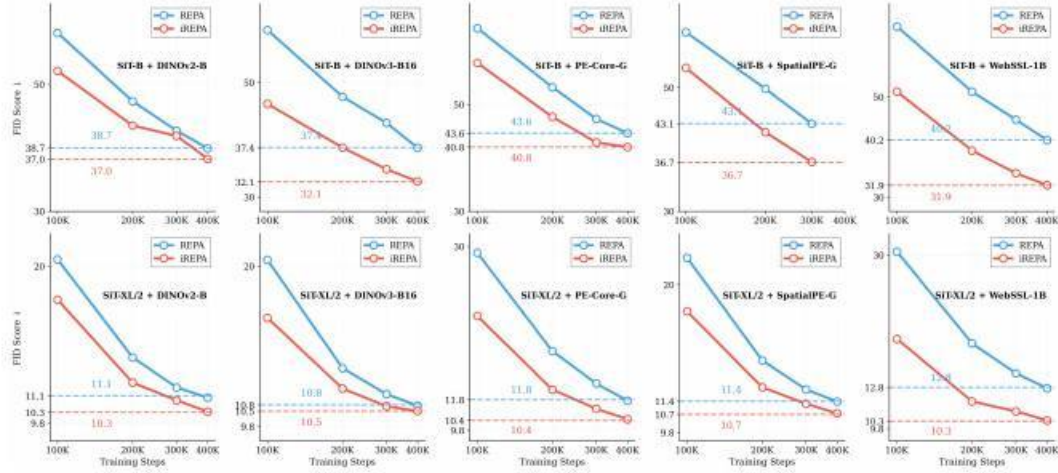


Figure 25: **Convergence results at 512×512 resolution.** Accentuating spatial features consistently improves convergence speed at both resolutions we test—ImageNet-256 (Figure 7) and ImageNet-512 (above).

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. To probe generalization beyond ImageNet-style class-conditional generation, we further evaluate the proposed spatial improvements on multimodal text-to-image (T2I) tasks. Following REPA (Yu et al., 2024), we adopt MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with a range of pretrained vision encoders (DINOv2, CLIP, WebSSL, PE, etc.). As in REPA (Yu et al., 2024), all models are trained for 150K steps with a batch size of 256 and evaluated with an ODE sampler at 50 NFE. Fig. 26 compares REPA before and after the spatial improvements (iREPA). In this comparatively limited T2I study, the improvements again transfer, delivering consistent convergence gains across every pretrained encoder considered (DINOv2, CLIP, WebSSL, PE, etc.); a broader evaluation of T2I generation remains an avenue for future work.

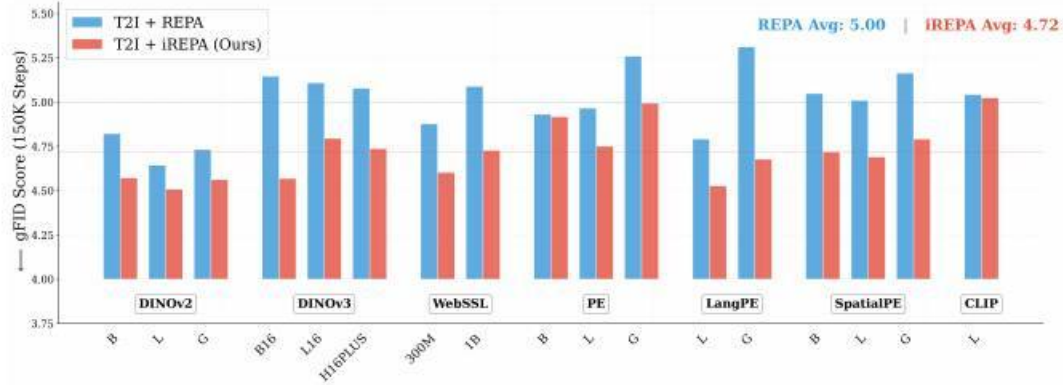


Figure 26: **Text-to-image generation across encoder variants.** Accentuating the transfer of spatial features from the target representation to the diffusion features consistently improves convergence speed in both the ImageNet setting (see §4) and this multimodal T2I study (above), with gains across every pretrained encoder considered (DINOv2, CLIP, WebSSL, PE, etc.).

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOV3-L AND MAE-L)

To confirm that our conclusions are not driven by outliers, we repeat the correlation analysis of Section 3 after removing MoCOV3-L and MAE-L, the two encoders identified as outliers. Figure 28 reports the updated correlations across model sizes. Spatial structure continues to correlate far more strongly with generation performance than linear probing accuracy does. Notably, once the outliers are removed, linear probing exhibits a small *positive* correlation with gFID—that is, higher linear probing performance is associated with *worse* generation. This is precisely consistent with Sec. 2, in which target representations with higher global semantic performance (linear probing accuracy) frequently yield similar or worse generation under representation alignment (REPA).

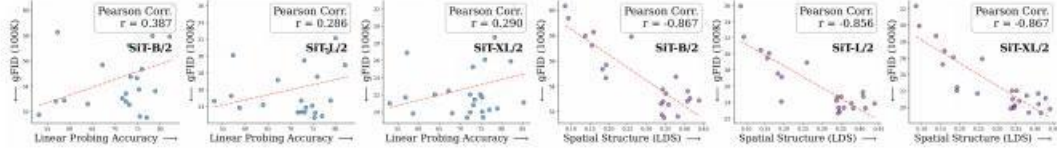


Figure 27: Correlation analysis without outliers. After removing the outliers (MoCOV3-L and MAE-L), spatial structure retains a far stronger correlation with generation performance (Pearson $|r| > 0.85$) than linear probing accuracy (Pearson $|r| < 0.38$) at all model sizes (B, L, XL). Notably, with the outliers removed, linear probing shows a small *positive* correlation with gFID—higher linear probing performance is associated with *worse* generation. This is consistent with Sec. 2, in which target representations with higher global semantic performance (linear probing accuracy) frequently yield similar or worse generation under representation alignment (REPA).

M.4 ADDITIONAL QUALITATIVE RESULTS

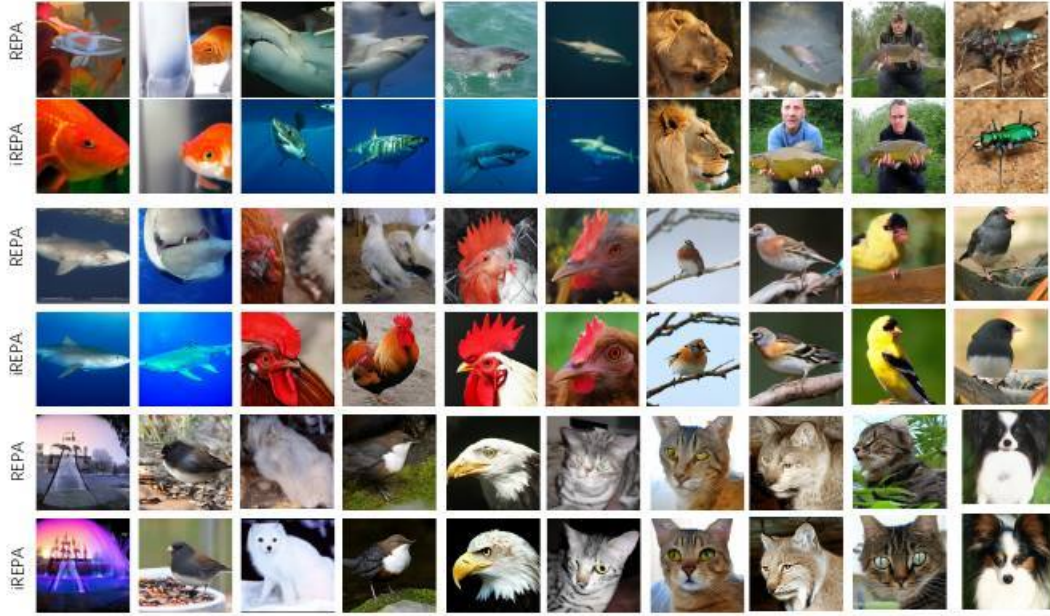


Figure 28: **Additional Qualitative Results** comparing generation outputs before and after the spatial improvements with REPA. All results use PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs), and a classifier-free guidance scale of 4.0. Consistent with the quantitative improvements (§4), the spatial improvements (iREPA) visibly enhance the visual quality and coherence of the generated outputs.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

Beyond the traditional evaluation metrics (Inception Score, FID, sFID, Precision, Recall), we further assess the robustness of our findings using an alternative generation-quality metric (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). Specifically, we adopt the CMMD metric (Jayasumana et al., 2024) with the PyTorch implementation of (Paul, 2024), using the standard reference set from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) as reference images. We then re-examine the key findings of both §3 and §4 under CMMD.

Spatial structure correlates far more strongly with generation performance than linear probing.

To test the robustness of the analysis in §3, we repeat the large-scale correlation study across vision encoders using the CMMD metric (Jayasumana et al., 2024). Results appear in Figure 29 (SiT-B/2, 100K steps, REPA). Every spatial metric correlates strongly with generation performance under CMMD (Pearson $|r| > 0.88$), whereas linear probing remains essentially uncorrelated ($|r| = 0.074$)—indicating that the central finding of §3 is unchanged whether generation quality is measured by FID or CMMD.

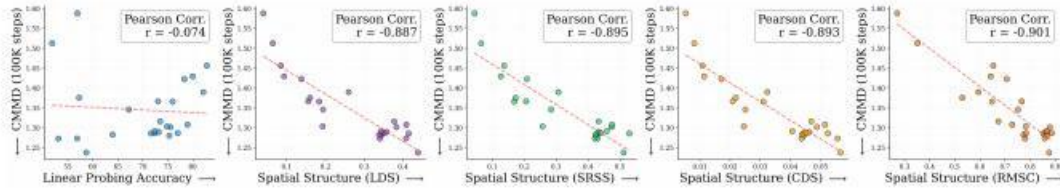


Figure 29: Correlation analysis with the CMMD metric. We repeat the correlation analysis of §3 using CMMD (Jayasumana et al., 2024) in place of gFID (SiT-B/2, 100K steps, REPA). Every spatial metric correlates strongly with generation performance (Pearson $|r| > 0.88$), whereas linear probing remains essentially uncorrelated ($|r| = 0.074$), indicating that the key finding of §3 is robust to the choice of evaluation metric.

Accentuating the transfer of spatial features improves generation quality. We likewise test the generality of the analysis in §4 under CMMD (Jayasumana et al., 2024), alongside the traditional metrics (gFID, sFID, IS, etc.). As Table 8 shows, for every vision encoder considered, accentuating spatial-feature transfer (iREPA) improves convergence speed under both CMMD and the traditional metrics (IS, FID, sFID, Prec., Recall), and the gains hold both with and without classifier-free guidance (CFG).

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Additional results with alternative evaluation metrics.** We report complementary results using the CMMD metric (Jayasumana et al., 2024), computed with the PyTorch implementation of (Paul, 2024) and the standard reference set from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021). All results use SiT-XL/2, batch size 256, and traditional REPA as the baseline; the iREPA rows are highlighted, and per-metric bests are in bold. For every vision encoder, accentuating spatial-feature transfer (iREPA) improves convergence speed under both CMMD and the traditional metrics (IS, FID, etc.), and the improvement holds both with and without classifier-free guidance, indicating that the key findings of §4 are robust to the choice of evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 examine the effect of spatial normalization (§4) on SAM2-S (46M) encoder features². As reported in Table 9, the spatial normalization layer still improves SAM2 performance, but the gains are comparatively marginal—which is exactly what the mechanism predicts and, in fact, provides a further consistency check on it. Spatial normalization operates by removing the global component (the mean of patch tokens) to sharpen spatial contrast; because SAM2 already carries little to no global information (validation accuracy < 24%), there is little global component to remove, and the achievable increase in spatial contrast is correspondingly small (see Figure 30).

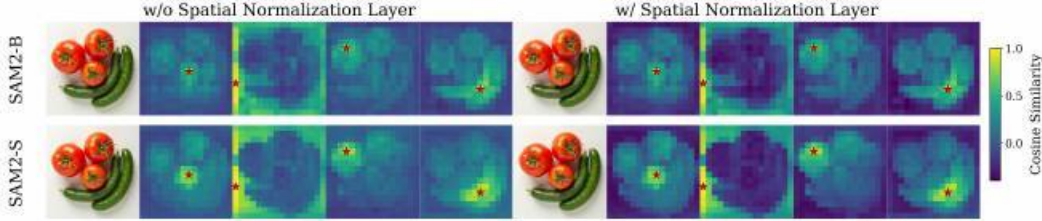


Figure 30: **Impact of spatial normalization on SAM2.** For SAM2, the spatial normalization layer does enhance spatial contrast, but the improvement is marginal. The reason is intrinsic to the mechanism: spatial normalization (§4) enhances spatial contrast by removing the global component (the mean of patch tokens). Since SAM2 already carries little to no global information (validation accuracy < 24%), there is little to remove, so spatial contrast increases only slightly.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **Impact of spatial improvements on SAM2.** iREPA still improves over the baseline, though only marginally—exactly as the mechanism predicts, since SAM2 carries little global component for spatial normalization to remove. All results use SiT-XL/2, without classifier-free guidance, SAM2-S (46M) as the vision encoder, and traditional REPA as the baseline; the iREPA rows are highlighted, and per-metric bests are in bold.

²As in (Bolya et al., 2025), we use the intermediate encoder output for SAM2 features rather than the mask logits: (Bolya et al., 2025) shows that although the mask logits yield sharper spatial maps, they are themselves unsuitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis of §3 using post-full-finetuning validation accuracy³ in place of linear probing. Results appear in Fig. 31 (SiT-XL/2, 100K steps, REPA); for the full-finetuning accuracy, we finetune the entire vision encoder with a learning rate of $5e-5$, a warmup ratio of 0.1, and 30 epochs. Every spatial metric again correlates far more strongly with generation performance than full-finetuning accuracy does. Strikingly, gFID exhibits a weak *positive* correlation with full-finetuning accuracy (Pearson $r = 0.317$): higher validation accuracy is associated with higher gFID and therefore *worse* generation. This reinforces the observations of §2, in which target representations with higher global semantic performance frequently yield similar or worse generation under representation alignment (REPA).

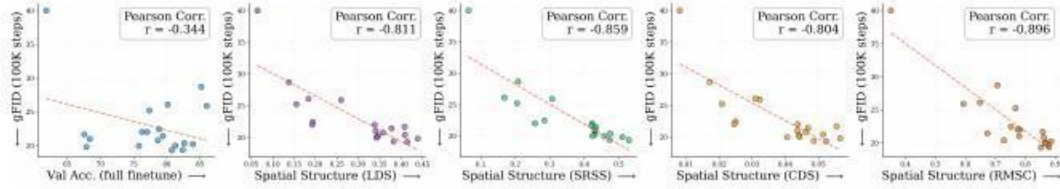


Figure 31: Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis of §3 using post-full-finetuning validation accuracy in place of linear probing (SiT-XL/2, 100K steps, REPA). Every spatial metric correlates far more strongly with generation performance than full-finetuning accuracy. Strikingly, gFID shows a weak *positive* correlation with full-finetuning accuracy (Pearson $r = 0.317$): higher validation accuracy is associated with higher gFID, i.e., worse generation. This is consistent with §2, where target representations with higher global semantic performance frequently yield similar or worse generation under representation alignment (REPA).

³While the overall findings are unchanged, in the context of REPA linear probing is arguably the more faithful estimate of the global information in a “pretrained” vision encoder: REPA regularizes against the “pretrained” encoder features themselves, and finetuning the encoder can alter the amount of global information. Following REPA (Yu et al., 2024), we therefore retain linear probing accuracy as the default measure of global information.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Generation performance of pixel-space diffusion models.** Despite its simplicity, iREPA combined with JiT (Li & He, 2025a) surpasses recent competitive pixel-space diffusion methods, including DeCo (Ma et al., 2025b), at matched parameter count. All results use 200K training iterations with a batch size of 256, evaluated via 50-step Euler sampling without classifier-free guidance; DeCo results are taken directly from (Ma et al., 2025b). The JiT+iREPA row is highlighted, and per-metric bests among the compared methods are in bold.